RNA-seq (3 batches)
Raw QC & trimming (FastQC / fastp / MultiQC)
Alignment & gene counts (STAR + featureCounts)
Merged counts + batch-aware DE (DESeq2, design = ~ batch + group)
Per-gene log2FC (per batch)
Mechanism gene-set Z (axis-level)
Failure-mode QC & dedup (GCguard filtering)
Within-batch PON normalization (GC median profile)
Mechanistic indicators: CLPS, MLCL, ES, oxCL/CL, ECL, EPG
Batch-specific axis Z (lipid mechanistic axis scores)
HO vs WT axis-level effects (δ_axis_z, lipid_axis)
Axis-level multi-omics integration (Stouffer 等权方法，用于 per-axis multi-omics Z)
Upstream & downstream RNA pathway Z (上游信号通路 + EphB1 下游基因集)
Axis–pathway alignment (axis × pathway 对齐度 mean_Z, cor, support_Z)
Sample-level RNA-lipid scores (EphB1 下游 M_score_RNA，指向 Y_score_RNA / Y_score_lipid)
Mediation analysis (X = group, M = RNA, Y = lipid / axis)
Mechanistic axis scores (RNA & lipid, per batch)